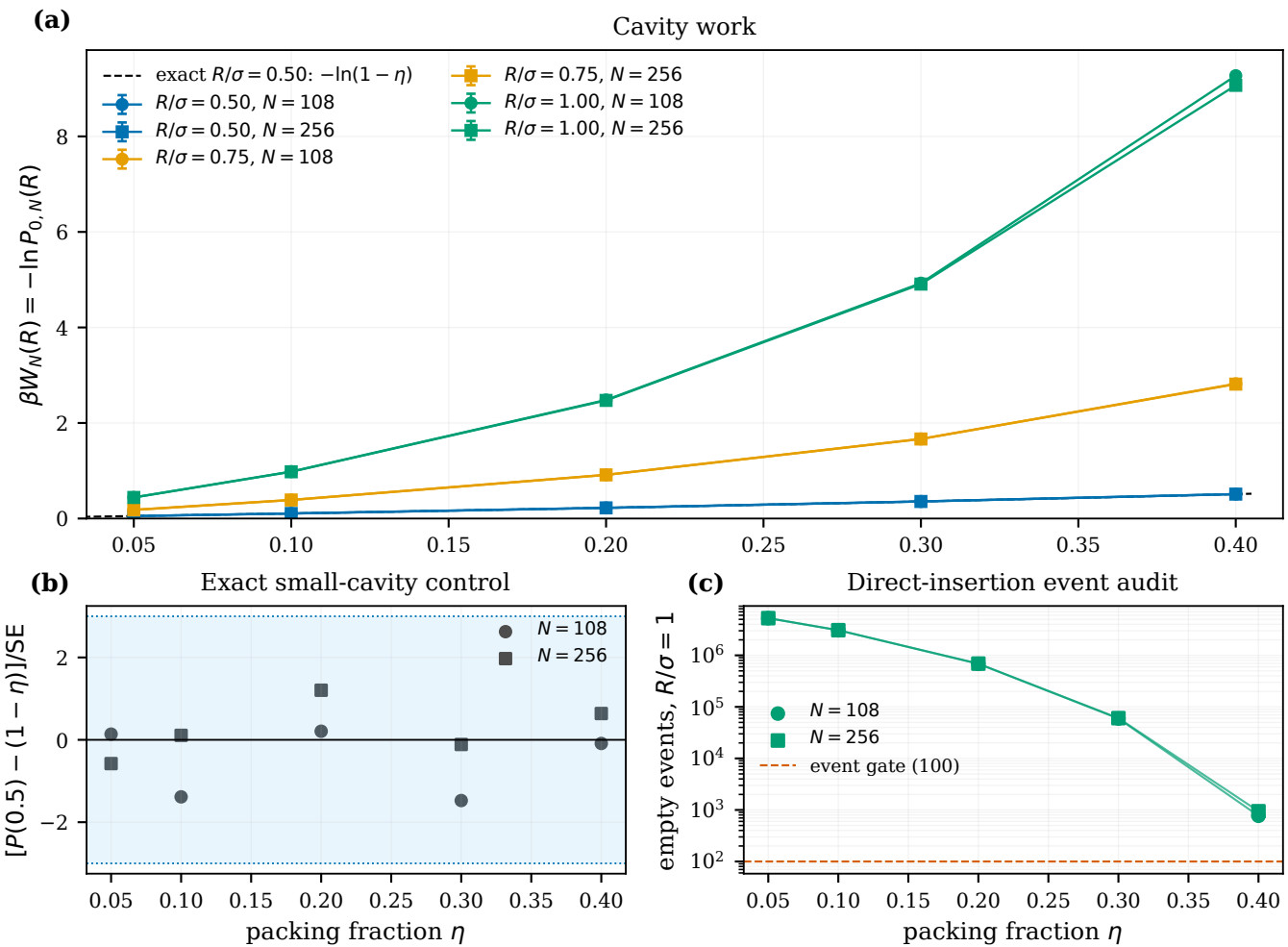
(a)
Cavity work
exact $R/\sigma = 0.50$: $-\ln(1-\eta)$
$R/\sigma = 0.50$, $N = 108$
$R/\sigma = 0.50$, $N = 256$
$R/\sigma = 0.75$, $N = 108$
$R/\sigma = 0.75$, $N = 256$
$R/\sigma = 1.00$, $N = 108$
$R/\sigma = 1.00$, $N = 256$
$\beta W_N(R) = -\ln P_{0,N}(R)$
0
2
4
6
8
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
(b)
Exact small-cavity control
$N = 108$
$N = 256$
$[P(0.5) - (1-\eta)]/\mathrm{SE}$
−2
0
2
packing fraction η
(c)
Direct-insertion event audit
$N = 108$
$N = 256$
event gate (100)
empty events, $R/\sigma = 1$
10^2
10^3
10^4
10^5
10^6
packing fraction η